Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Moonsub Shim	Email	mshim@illinois.edu
Institution / Department	Department of Materials Science & Engineering, UIUC			
Address	1304 W. Green St. Urbana, IL 61801 USA			

2. Technology Overview

Title of Proposed Technology	*In Situ* Optical Monitoring of Colloidal Quantum Dot Growth for Precision and Reproducible Synthesis and their Application in QD LEDs			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Electrical/Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	*Nano-Application Technologies*	*Display*	*Display Components and Materials*
Concept of Proposed Technology	The PI's group has recently developed a high-temperature, vacuum tolerant *in situ* optical probe that can measure UV-Vis and PL spectra in harsh chemical environments at > 400°C with temporal resolution of ~ms (currently limited by the speed of the commercial spectrometer used and therefore may be improved to ~μs or better with a faster spectrometer).[1] The proposed work would build on this system and leverage the PI's expertise in QD synthesis and characterization[2-5] to develop *in situ* probes that can lead to precision, reproducibility, and efficiency in QD production. The efforts of this project would involve adapting the current laboratory scale probe to be compatible with industrial processes and developing analysis techniques that will allow conversion of spectral data into physical parameters, including size, size distribution and concentration, such that these parameters can be followed in real time during production. These efforts would lead to improved reproducibility and facilitate synthesis of high-quality QD materials that are used for applications such as self-emissive QD LEDs, solar cells, detectors and beyond. Furthermore, reaction scaling and development of new or less-explored compositions may be accelerated by incorporating our *in situ* approach.			

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Rationale for Joint R&D Project	*In situ* optical probe can provide critical information about QD growth, which in turn can help to improve the quality of materials, accelerate the development of reproducible synthesis, and enable scale-up. The ability to monitor the evolution of size, size distribution and concentration of QDs in real time during synthesis can help to elucidate growth mechanisms and allow identification of key factors such as causes of size broadening and reproducibility issues. These factors are arguably the most critical outstanding issues in the production of QDs for a variety of technologies including emissive displays and photodetectors. The ability to monitor these critical parameters in real time means that each synthesis can be tuned to desired product specifications much more precisely and efficiently, leading to improved quality and reduced waste. Furthermore, developing different compositions as well as heterostructures such core/shell structures may be accelerated through this ability.
Keywords	1. Colloidal Quantum Dots and Heterostructures 2. High-Temperature *In Situ* Optical Probe 3. Growth Kinetics and Mechanisms 4. Self-Emissive QD LEDs 5.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)

R&D Objectives

High reaction temperatures are inaccessible with commercial optical "dip" probes, which have a hard ceiling well below the reaction temperature for synthesizing high-quality QD materials.[6-8] The PI's group has developed an *in situ* optical probe that can measure UV-Vis and PL spectra in harsh chemical environments and under vacuum conditions at > 400°C with fast time response (**Fig. 1a**). This capability provides an exciting opportunity to examine nucleation and growth kinetics and to gain insights into key synthesis parameters directly at conditions relevant to QD production. Such knowledge is essential for developing precise, reproducible, and efficient manufacturing of high-performance materials. The objectives of this project are to 1) improve the probe design and modify it to allow compatibility for *in situ* monitoring of QD synthesis in industry-relevant conditions, 2) develop analysis approach that will allow real-time tracking of critical parameters of size, size distribution, and concentration, and 3) demonstrate applicability to new compositions/heterostructures.

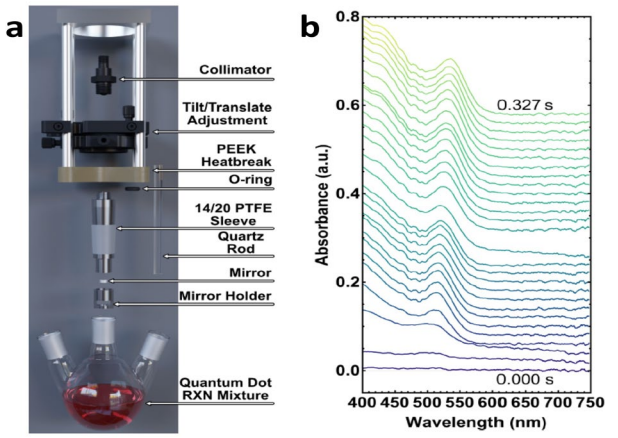

Fig. 1. (**a**) Robust high-temperature vacuum-tolerant *in situ* UV-Vis probe. (**b**) Example of *in situ* absorption spectra (offset for clarity) of CdSe QD reaction mixture with rapid acquisition at the beginning of reaction at 380°C.

R&D Target Specifications

The current optical probe's adapter for fiber optics is rather bulky and therefore the first goal of this project is to miniaturize this component to allow versatility in its use and minimize the need for manual alignment of optics. In order to convert spectral information to physical quantities (size, size distribution, and concentration), we will build on the PI's group's advances in developing an automated analysis approach using Statistically-sensitive Nonlinear Iterative Peak clipping to accurately analyze these large datasets rapidly.[1] The resulting information will lead to new insights on growth mechanisms and pave a path to precise, reproducible and efficient synthesis of QD materials with high quality, especially with respect to their optical properties. If suitable SME partner can be found, we will also explore LEDs or related device application of the resulting materials.

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R&D Plan

The current probe's temporal resolution of ~10 ms is limited by the data-saving rate of the commercial spectrometer used, and an improvement of 4 orders of magnitude (~μs) with a faster commercial spectrometer can be anticipated. Even at 10-ms resolution, important information, such as initial nuclei size and temporal evolution of size and concentration (and therefore growth kinetics), may already be gained. Preliminary results of UV/Vis spectra acquired during the first 0.33 s of CdSe QD synthesis at 380°C are shown in **Fig. 1b**. Note that this set of spectra is only a tiny fraction of tens of thousands of spectra that can be collected during a single synthesis. The absorption peak (between 500 and 550 nm for the shown spectra set), once calibrated for the temperature dependent band gap, should allow determination of how size evolves with time. Its redshift can reveal growth kinetics, and its linewidth can provide temporal evolution of size distribution. Optical density change at shorter wavelengths (e.g., ~400 nm) should scale with the concentration of the QDs in the reaction mixture. We will build on these accomplishments to establish sizing curves at reaction temperatures, enabling real-time tracking of size evolution. Combined with temporal dependence of size distribution and concentration from these experiments, we aim to provide new insights on the growth mechanism and demonstrate generality of our approach to developing precise, reproducible and efficient synthesis that can be applied to different compositions.

In addition to examining the model QDs, we will extend our *in situ* studies to solution heteroepitaxial growth. For example, we will explore multi-quantum dot nanorods (*n*-DNRs, where *n* represents the number of Dots in a NanoRod) that the PI's group has recently developed (**Fig. 2a**).[9] *n*-DNRs represent a new class of materials where the number of QDs and the interaction between them can be controlled through accessible solution synthesis. Preliminary efforts have led to the discovery of emission color change upon charge transfer via photochemical doping (**Fig. 2b**). This effect may lead to practical use if the charge injection is carried out electrically within a device setting. With the ability to tune each QD's size, shape and composition, the *n*-DNRs may represent a new platform to explore quantum systems such as the double QDs for quantum information science as well as to develop optical downconverters and self-emissive LED materials. These efforts will leverage the PI's expertise in QD LEDs and related devices.[10-13] Extending our *in situ* studies on solution heteroepitaxy will lead to critical information necessary for precision and reproducibility in the synthesis of these emerging materials.

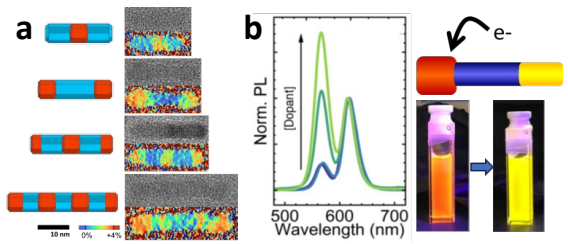

Fig. 2. (a) TEM images and corresponding strain maps and schematics of *n*-DNR of consisting of 1 through 4 QDs (top to bottom). (**b**) Evolution of PL of 2-DNR solution upon photochemical doping with electrons, where color switching can be readily observed by eye in the photographs.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The PI's research group will lead the efforts on the development of the *in situ* probe and automated analysis and characterize the resulting products using a wide range of facilities available at the University of Illinois. We will work closely with our SME partners to design *in situ* probe to be compatible with industry needs, improve/optimize QD production based on *in situ* studies, and explore resulting materials for potential applications in lighting and displays.
Expected R&R of Korean SME Partner	There are two main collaborative directions this research can take. The first is in the synthesis/production of QDs. The PI's group would seek input from SME partner on the industrial needs and requirements such that we can collaboratively design and develop *in situ* probe that meets relevant specifications to improve product quality and reproducibility. The second direction would be to explore the resulting high-quality QD materials for proof-of-concept applications in LEDs and related areas. In addition to synthesis and characterization of QDs, the PI has expertise in QD LEDs, solar cells and related device applications. We would work collaboratively with an SME partner interested in employing materials synthesized by the PI's group in similar applications – e.g., to develop proof-of-concept self-emissive displays based on QDs and related materials.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	We anticipate versatile *in situ* optical probe system with rapid, automated analysis approach that can be employed in industry-relevant conditions for QD synthesis. We aim to demonstrate improved precision, reproducibility, and efficiency in QD synthesis. The resulting QD-based materials may lead to improvements in self-emissive LEDs, photodetectors, and related applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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